# BCNF Proofs — MediNexus Schema

Database Design · Relational Schema Analysis

**Definition (BCNF).** A relation schema $R$ is in **Boyce–Codd Normal Form (BCNF)** if and only if for every non-trivial functional dependency $X \to Y$ that holds in $R$, $X$ is a superkey of $R$.

The MediNexus schema contains 19 relations. Each is analysed below by listing its functional dependencies, identifying all candidate keys, and verifying the BCNF condition.

## Contents

## 1. User

### Schema

User(USER_ID, PINCODE(FK), EMAIL, PASSWORD, REG_DATE, TYPE)

### Functional Dependencies

**FD1.** USER_ID → PINCODE, EMAIL, PASSWORD, REG_DATE, TYPE

### Candidate Keys

{USER_ID}

### BCNF Proof

1. The only non-trivial FD is FD1: USER_ID → all other attributes.
2. USER_ID is the sole candidate key and hence a superkey.
3. Every non-trivial FD has a superkey on its left-hand side.

∴ *User is in BCNF.*

## 2. Patients

### Schema

Patients(PATIENT_ID, NAME, DOB, GENDER, BLOOD_GROUP, EMG_CONTACT)

### Functional Dependencies

**FD1.** PATIENT_ID → NAME, DOB, GENDER, BLOOD_GROUP, EMG_CONTACT

### Candidate Keys

{PATIENT_ID}

### BCNF Proof

1. FD1 is the only non-trivial FD.
2. PATIENT_ID is the sole candidate key.
3. Its LHS is a superkey.

∴ *Patients is in BCNF.*

## 3. Doctor

### Schema

Doctor(DOC_ID, NAME, LICENSE_NO, SPECIALIZATION, EXPERIENCE, FEES, HOSPITAL_AFFILIATION)

### Functional Dependencies

**FD1.** DOC_ID → NAME, LICENSE_NO, SPECIALIZATION, EXPERIENCE, FEES, HOSPITAL_AFFILIATIO

**FD2.** LICENSE_NO → DOC_ID, NAME, SPECIALIZATION, EXPERIENCE, FEES, HOSPITAL_AFFILIATIO

**Candidate Keys**

{`DOC_ID`}, {`LICENSE_NO`}

**BCNF Proof**

1. FD1: `DOC_ID` is a candidate key ⇒ superkey. ✓
2. FD2: A medical licence uniquely identifies one doctor, so `LICENSE_NO` is also a candidate key ⇒ superkey. ✓
3. All non-trivial FD left-hand sides are superkeys.

∴ *Doctor is in BCNF.*

## 4. Location

**Schema**

`Location(PINCODE, CITY, STATE)`

**Functional Dependencies**

**FD1.** `PINCODE` → `CITY, STATE`

**Candidate Keys**

{`PINCODE`}

**BCNF Proof**

1. `PINCODE` → `CITY, STATE` is the only non-trivial FD.
2. `PINCODE` is the sole candidate key and is therefore a superkey.

∴ *Location is in BCNF.*

## 5. Visit_Time

**Schema**

`Visit_Time(SLOT_ID, DOC_ID(FK), START_TIME, END_TIME, MODE, DAY)`

**Functional Dependencies**

**FD1.** `SLOT_ID` → `DOC_ID, START_TIME, END_TIME, MODE, DAY`

**FD2.** `DOC_ID, DAY, START_TIME` → `SLOT_ID, END_TIME, MODE`

**Candidate Keys**

{`SLOT_ID`}, {`DOC_ID, DAY, START_TIME`}

**BCNF Proof**

1. FD1: `SLOT_ID` is a candidate key ⇒ superkey. ✓
2. FD2: The composite {`DOC_ID, DAY, START_TIME`} uniquely identifies a time slot, making it also a candidate key ⇒ superkey. ✓

∴ *Visit_Time is in BCNF.*

## 6. Appointment

### Schema

`Appointment(APP_ID, PATIENT_ID(FK), DOC_ID(FK), TIMESTAMP, DATE, STATUS)`

### Functional Dependencies

**FD1.** `APP_ID` → `PATIENT_ID, DOC_ID, TIMESTAMP, DATE, STATUS`

### Candidate Keys

{`APP_ID`}

### BCNF Proof

1. FD1 is the only non-trivial FD and its LHS `APP_ID` is the candidate key.

∴ *Appointment is in BCNF.*

## 7. Consultation

### Schema

`Consultation(CONSULT_ID, APP_ID(FK), DIAGNOSIS, SUMMARY, TIMESTAMP)`

### Functional Dependencies

**FD1.** `CONSULT_ID` → `APP_ID, DIAGNOSIS, SUMMARY, TIMESTAMP`

**FD2.** `APP_ID` → `CONSULT_ID, DIAGNOSIS, SUMMARY, TIMESTAMP`

### Candidate Keys

{`CONSULT_ID`}, {`APP_ID`}

### BCNF Proof

1. Each appointment produces at most one consultation, establishing a 1-to-1 mapping.
2. FD1: `CONSULT_ID` is a candidate key ⇒ superkey. ✓
3. FD2: `APP_ID` is also a candidate key ⇒ superkey. ✓

∴ *Consultation is in BCNF.*

## 8. Symptoms

### Schema

`Symptoms(TYPE, CONSULT_ID(FK))`

### Functional Dependencies

No non-trivial functional dependencies beyond the composite primary key.

### Candidate Keys

{`TYPE, CONSULT_ID`}

### BCNF Proof

1. The composite {`TYPE, CONSULT_ID`} is the sole candidate key.

2. There are no non-trivial FDs; the BCNF condition holds vacuously.

∴ *Symptoms is in BCNF.*

## 9. Vital_Log

### Schema

`Vital_Log(LOG_ID, PATIENT_ID(FK), TIMESTAMP, METRIC_TYPE, VALUE, UNIT)`

### Functional Dependencies

**FD1.** `LOG_ID` → `PATIENT_ID, TIMESTAMP, METRIC_TYPE, VALUE, UNIT`

### Candidate Keys

{`LOG_ID`}

### BCNF Proof

1. FD1 is the only non-trivial FD; its LHS is the primary key.

∴ *Vital_Log is in BCNF.*

## 10. Prescription

### Schema

`Prescription(PRES_ID, CONSULT_ID(FK), ISSUE_DATE)`

### Functional Dependencies

**FD1.** `PRES_ID` → `CONSULT_ID, ISSUE_DATE`

**FD2.** `CONSULT_ID` → `PRES_ID, ISSUE_DATE`

### Candidate Keys

{`PRES_ID`}, {`CONSULT_ID`}

### BCNF Proof

1. One consultation yields exactly one prescription (1-to-1).
2. FD1 LHS `PRES_ID` is a candidate key ⇒ superkey. ✓
3. FD2 LHS `CONSULT_ID` is a candidate key ⇒ superkey. ✓

∴ *Prescription is in BCNF.*

## 11. Prescription_Item

### Schema

`Prescription_Item(PRES_ID(FK), MED_CODE(FK), DOSAGE, FREQUENCY, QTY)`

### Functional Dependencies

**FD1.** `PRES_ID, MED_CODE` → `DOSAGE, FREQUENCY, QTY`

**Candidate Keys**

{PRES_ID, MED_CODE}

**BCNF Proof**

1. The composite {PRES_ID, MED_CODE} is the only candidate key.
2. No attribute in {DOSAGE, FREQUENCY, QTY} determines any other attribute, so FD1 is the only non-trivial FD.
3. Its LHS is a superkey.

∴ *Prescription_Item is in BCNF.*

## 12. Medicine

**Schema**

`Medicine(MED_CODE, NAME, BRAND_NAME, CATEGORY, DESCRIPTION)`

**Functional Dependencies**

**FD1.** MED_CODE → NAME, BRAND_NAME, CATEGORY, DESCRIPTION

**Candidate Keys**

{MED_CODE}

**BCNF Proof**

1. FD1 is the only non-trivial FD; its LHS is the primary key.

∴ *Medicine is in BCNF.*

## 13. Pharmacy

**Schema**

`Pharmacy(PHARMACY_ID, TAX_ID, CONTACT_NO, P_NAME)`

**Functional Dependencies**

**FD1.** PHARMACY_ID → TAX_ID, CONTACT_NO, P_NAME

**FD2.** TAX_ID → PHARMACY_ID, CONTACT_NO, P_NAME

**Candidate Keys**

{PHARMACY_ID}, {TAX_ID}

**BCNF Proof**

1. Each pharmacy has a unique tax registration number.
2. FD1 LHS PHARMACY_ID is a candidate key ⇒ superkey. ✓
3. FD2 LHS TAX_ID is a candidate key ⇒ superkey. ✓

∴ *Pharmacy is in BCNF.*

## 14. Stocks

### Schema

Stocks(PHARMACY_ID(FK), MED_CODE(FK), UNIT_PRICE, QTY)

### Functional Dependencies

**FD1.** PHARMACY_ID, MED_CODE $\rightarrow$ UNIT_PRICE, QTY

### Candidate Keys

{PHARMACY_ID, MED_CODE}

### BCNF Proof

1. The composite {PHARMACY_ID, MED_CODE} is the only candidate key.
2. FD1 is the only non-trivial FD; its LHS is a superkey.

$\therefore$ *Stocks is in BCNF.*

## 15. Bill

### Schema

Bill(BILL_ID, PATIENT_ID(FK), DATE, TOTAL_AMOUNT)

### Functional Dependencies

**FD1.** BILL_ID $\rightarrow$ PATIENT_ID, DATE, TOTAL_AMOUNT

### Candidate Keys

{BILL_ID}

### BCNF Proof

1. FD1 is the only non-trivial FD; its LHS is the primary key.

$\therefore$ *Bill is in BCNF.*

## 16. Consultation_Bill

### Schema

Consultation_Bill(CONSULT_ID(FK), BILL_ID(FK/PK))

*Note:* CONSULT_ID is the primary key of this relation (inherited from Consultation). BILL_ID is a foreign key referencing Bill. This is a 1-to-1 bridge; CONSULT_ID uniquely identifies the row.

### Functional Dependencies

**FD1.** CONSULT_ID $\rightarrow$ BILL_ID (CONSULT_ID is PK; each consultation maps to one bill)

**FD2.** BILL_ID $\rightarrow$ CONSULT_ID (BILL_ID is unique here; each bill maps to one consultation)

### Candidate Keys

{CONSULT_ID} (primary key); {BILL_ID} (alternate key, unique constraint)

### BCNF Proof

1. `CONSULT_ID` is the designated primary key and a superkey.
2. FD1 LHS `CONSULT_ID` is a superkey. ✓
3. FD2 LHS `BILL_ID` is an alternate candidate key (unique per row) ⇒ also a superkey. ✓
4. All non-trivial FD left-hand sides are superkeys.

∴ *Consultation_Bill is in BCNF.*

## 17. PHARMA_BILL

### Schema

`PHARMA_BILL(BILL_ID(FK/PK), PHARMACY_ID(FK))`

*Note:* `BILL_ID` is the primary key of this relation (each pharma bill entry is uniquely identified by the bill). `PHARMACY_ID` is a foreign key referencing Pharmacy, indicating which pharmacy issued the bill.

### Functional Dependencies

**FD1.** `BILL_ID` → `PHARMACY_ID` (`BILL_ID` is PK; each bill is associated with exactly one pharmacy)

**FD2.** `PHARMACY_ID` → `BILL_ID` only if a pharmacy issues exactly one pharma bill — this is **not** assumed in general, so this FD does **not** hold.

### Candidate Keys

{`BILL_ID`}

### BCNF Proof

1. `BILL_ID` is the sole candidate key (a pharmacy can have multiple bills; a bill belongs to one pharmacy).
2. The only non-trivial FD is FD1: `BILL_ID` → `PHARMACY_ID`.
3. Its LHS `BILL_ID` is the primary key ⇒ superkey. ✓

∴ *PHARMA_BILL is in BCNF.*

## 18. Pharma_Bill_Item

### Schema

`Pharma_Bill_Item(BILL_ID(FK), MED_CODE(FK), QTY, UNIT_PRICE)`

### Functional Dependencies

**FD1.** `BILL_ID, MED_CODE` → `QTY, UNIT_PRICE`

### Candidate Keys

{`BILL_ID, MED_CODE`}

### BCNF Proof

1. The composite {`BILL_ID`, `MED_CODE`} is the only candidate key.
2. FD1 is the only non-trivial FD; its LHS is a superkey.

∴ *Pharma_Bill_Item is in BCNF.*

## 19. Payment

### Schema

`Payment(TRANSACTION_ID, BILL_ID(FK), GATEWAY, MODE, REF_NO)`

### Functional Dependencies

**FD1.** `TRANSACTION_ID` → `BILL_ID, GATEWAY, MODE, REF_NO`

**FD2.** `BILL_ID` → `TRANSACTION_ID, GATEWAY, MODE, REF_NO`

### Candidate Keys

{`TRANSACTION_ID`}, {`BILL_ID`}

### BCNF Proof

1. Each bill is settled by exactly one payment (1-to-1).
2. FD1 LHS `TRANSACTION_ID` is a candidate key ⇒ superkey. ✓
3. FD2 LHS `BILL_ID` is a candidate key ⇒ superkey. ✓

∴ *Payment is in BCNF.*

## Summary

| # | Relation | Candidate Key(s) | Non-trivial FDs | BCNF |
|---|---|---|---|---|
| 1 | User | `USER_ID` | 1 | ✓ |
| 2 | Patients | `PATIENT_ID` | 1 | ✓ |
| 3 | Doctor | `DOC_ID`, `LICENSE_NO` | 2 | ✓ |
| 4 | Location | `PINCODE` | 1 | ✓ |
| 5 | Visit_Time | `SLOT_ID`; (`DOC_ID, DAY, START_TIME`) | 2 | ✓ |
| 6 | Appointment | `APP_ID` | 1 | ✓ |
| 7 | Consultation | `CONSULT_ID`, `APP_ID` | 2 | ✓ |
| 8 | Symptoms | (`TYPE, CONSULT_ID`) | 0 | ✓ |
| 9 | Vital_Log | `LOG_ID` | 1 | ✓ |
| 10 | Prescription | `PRES_ID`, `CONSULT_ID` | 2 | ✓ |
| 11 | Prescription_Item | (`PRES_ID, MED_CODE`) | 1 | ✓ |
| 12 | Medicine | `MED_CODE` | 1 | ✓ |
| 13 | Pharmacy | `PHARMACY_ID`, `TAX_ID` | 2 | ✓ |
| 14 | Stocks | (`PHARMACY_ID, MED_CODE`) | 1 | ✓ |
| 15 | Bill | `BILL_ID` | 1 | ✓ |
| 16 | Consultation_Bill | `CONSULT_ID` (PK); `BILL_ID` (alt.) | 2 | ✓ |
| 17 | PHARMA_BILL | `BILL_ID` (PK) | 1 | ✓ |
| 18 | Pharma_Bill_Item | (`BILL_ID, MED_CODE`) | 1 | ✓ |
| 19 | Payment | `TRANSACTION_ID`, `BILL_ID` | 2 | ✓ |

**Conclusion.** All 19 relations in the MediNexus relational schema satisfy the BCNF condition. In every case, the left-hand side of each non-trivial functional dependency is either the primary key or an alternate candidate key, and is therefore a superkey.